

AI-Based Bankruptcy Prediction Using Financial Statements

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Abstract:

Corporate bankruptcy prediction is a critical component of financial risk management; however, traditional statistical models often struggle to process the volume and non-linear complexity of modern financial data. This study investigates the efficacy of Artificial Intelligence (AI) and machine learning techniques in predicting corporate bankruptcy using data extracted directly from financial statements. By evaluating key profitability, liquidity, leverage, and efficiency ratios, the research compares the predictive performance of AI-based models—such as decision trees and advanced classification algorithms—against traditional econometric approaches. The methodology involves training, testing, and cross-validating these models on historical financial data to identify hidden relational structures associated with financial distress. The empirical findings demonstrate that AI-based models significantly outperform traditional methods in predictive accuracy, adaptability, and the dynamic weighting of critical variables, with profitability and liquidity ratios emerging as the strongest indicators of financial stability. While the study highlights the transformative potential of AI as a proactive early warning system, it also addresses inherent challenges, including the "black box" nature of machine learning interpretability and a heavy reliance on pristine data quality. Ultimately, this research concludes that integrating AI into financial analysis fundamentally enhances corporate risk assessment, providing investors, auditors, and policymakers with a highly effective tool for safeguarding financial stability.

Keywords:

Bankruptcy Prediction, Artificial Intelligence (AI), Machine Learning, Financial Risk Management, Financial Ratios, Corporate Distress, Predictive Analytics

Introduction: "The What & Why"

In today's rapidly evolving economic environment, businesses operate under high financial pressure driven by technological disruption, and unexpected macroeconomic shocks. Corporate bankruptcy is a repeated event that affects all the concerned ones in the firm. The failure of most firms has demonstrated how financial disasters can spread quickly and create systematic risk and economy will not be stable anymore.

Therefore, it's very important to predict bankruptcy before it occurs. This has become one of the most challenging topics in modern financial analysis. Traditional bankruptcy prediction methods involved financial ratio analysis and statistical models like Z-score (Altman) and logistic regression. These models apply data Fuente from financial statements (ratios based on profitability, liquidity, leverage, and efficiency) to predict the probability of firm success or failure. Though they have provided valuable insight for decades, these methods typically find their predictions limited by assumptions of linearity, static relationships, with data processing capacities constrained. As the financial data become increasingly large and intricate, these traditional models cannot effectively capture nonlinearities or hidden relational structures existing in financial information. Artificial intelligence, particularly machine learning techniques, represents a transformative development in financial forecasting and risk assessment. Unlike traditional statistical methods, AI-base models can process large volumes of structured financial data, detect subtle relationships, adapt to new information, and improve predictive accuracy over time.

By applying algorithms such as decision trees... researchers can potentially enhance early warning systems for corporate distress. AI does not merely automate traditional analysis; it redefines how financial risk is identified and measured. Despite the growing interest in AI applications in finance, questions remain regarding its effectiveness and usefulness in bankruptcy prediction. Can AI models outperform traditional statistical approaches? Which financial statement variables contribute to prediction accuracy? The most important question is how can these models be applied in real world decision-making by auditors, banks, policymakers, and investors?

This study seeks to put these questions under consideration by examining AI-based bankruptcy prediction using financial statement data. The most important objective is to evaluate if AI techniques improve predictive performance compared to traditional models and to assess their practical implications. This research aims to:

Identify key financial ratios derived from financial statements that are relevant to bankruptcy prediction.

Develop AI-based models to estimate the profitability of corporate failure.

Compare the predictive accuracy of AI models with traditional statistical methods.

Analyse the relative importance of financial variables within AI algorithms.

Evaluate model performance across different time horizons and economic conditions.

Assess the robustness and stability of predictions using validation techniques.

Examine the practical implications of AI-based bankruptcy prediction for investors, financial institutions, auditors, and regulators.

By applying AI in this field, economic losses will be reduced, and the field of risk financial risk management will evolve. This is not only a technical innovation but also a tool for the enhancement of financial stability and decision-making process.

Literature Review: "The story so far "

The topic was chosen because of the growing significance of financial risk management and the quickly changing business landscape of today. The global economy has seen an increase in corporate failures, financial crises, and unanticipated bankruptcies over the past few decades, impacting both large multinational corporations and smaller businesses. These incidents have shown how susceptible businesses can be to changes in the economy, bad management choices, and outside shocks.

As a finance student, I see that we are frequently exposed to these relative realities through our coursework, case studies, and real-world examples. This exposure sparked our curiosity to better understand the mechanism behind financial failure and, more importantly, whether it is possible to anticipate such outcomes before they occur.

Furthermore, the effects of bankruptcy go well beyond the companies themselves. Employees run the risk of losing their jobs, creditors might not get their money back, investors might lose a lot of money, and the economy might be unstable.

These far-reaching consequences emphasize how important it is to have trustworthy instruments and techniques for early financial distress prediction. This inspired us to investigate how contemporary technological developments can improve conventional financial analysis.

Our choice was also greatly influenced by the quick advancement of artificial intelligence (AI) in few industries, most notably the financial sector (Beainy, 2026).

AI has completely changed how risks are evaluated, decisions are made, and data is analyzed. To increase accuracy and efficiency, financial institutions are depending more on data-driven models. Therefore, we can link theoretical knowledge with useful, real-world applications by looking at how AI can be used (Sawaya, et al, 2026) to predict bankruptcy. Because bankruptcy forecasts are so important to decision-making, this subject is especially significant in this new era, led by the United States equity market where every single change impact not only a specific company but the entire world (Kamel, Beainy & Bteish 2025). Investors rely on it to assess possible risks and steer clear of losing ventures. Before making loans, banks and other financial organizations used it to evaluate borrowers' creditworthiness. It is used by regulators and auditors to keep an eye on businesses' financial stability and guarantee openness.

Although traditional statistical models have been used extensively for this purpose, they frequently find it difficult to keep up with the increasing volume and complexity of financial data (jafarigorzin et Al., 2026). We hope to investigate whether these cutting-edge methods can offer more precise, effective, and timely insights by incorporating AI into bankruptcy prediction.

In the end, this subject is very pertinent since it shows how cutting-edge technologies can improve financial stability, lower economic losses, and facilitate better decision-making in a world that is becoming more complex.

Methodology: "The how we did it "

To conduct this study, we relied primarily on financial statement data as the primary source of information. Financial statements, including the balance sheet, income statements and cash flow statement provide detailed insights into a company's and that's just the performance and position. From these statements, we extracted key financial ratios that are commonly used in bankruptcy prediction. These include profitability ratios such as return on assets and net profit margin,

liquidity ratios such as current be sure and quick ratio, leverage ratios such as debt-to-equity ratio and efficiency ratios such as asset turnover point. These variables were carefully selected because they collectively provide a comprehensive overview of a company's financial condition. The methodology is based on the application of artificial intelligence techniques particularly machine learning models, to analyze the collected data and predict the probability of bankruptcy. The process begins with data collection from reliable sources, followed by data preparation. This stage involves cleaning the data set by removing inconsistencies, handling missing values and ensuring that all variables are properly formatted. Data processing is a crucial step, as the accuracy of the model largely depends on the quality of the data input. Once the data set is prepared, the selected financial variables are used as inputs for machine learning models. Among the techniques considered are the decision trees, logistic regression models enhanced with AI, and other classification algorithms. These models are trained using historical data, where the financial characteristics of companies are linked to known outcomes bankrupt or non-bankrupt. Through this training process, the models learned to identify patterns and relationships that are associated with financial distress.

After training, the models are tested on new data to evaluate their predictive performance. This involves comparing the predicted outcomes with the actual results to measure accuracy. Various evaluation metrics such as precision, recall, and overall accuracy are used to assess the effectiveness of the models.

Additionally, validation techniques, such as cross validation, are applied to ensure that the models are reliable and not overfitting the data. The results obtained from AI models are compared with those generated by traditional statistical methods.

This comparison allows us to determine whether AI provides a significant improvement in prediction accuracy and makes efficiency. By following this structured and systematic approach, the study aims to provide a clear assessment of the potential benefits and limitations of using AI and bankruptcy prediction.

Results: "The facts "

The findings of this study indicate that artificial intelligence-based models have strong potential in improving the accuracy and effectiveness of bankruptcy prediction. Compared to traditional statistical approaches, machine learning techniques demonstrate greater flexibility and the ability to capture complex, nonlinear relationships within financial data. This is particularly important given the increasing complexity of financial environments, where multiple variables interact in ways that are not always captured by a conventional model. The results showed that AI models generally outperform traditional methods in terms of predictive accuracy, especially when large data sets are used.

This improvement can be attributed to the ability of machine learning algorithms to process vast amounts of data and identify hidden patterns that may not be immediately visible through standard analysis. As a result, these models can provide earlier and more reliable warnings of potential financial distress.

Another important finding is that certain financial ratios play a more significant role in predicting bankruptcy than others. And profitability and liquidity ratios appear to be strong indicators of company's financial stability.

AI models can identify these key variables and assign them appropriate importance within the prediction process. This dynamic weighting enhances the overall performance of the model and allows for more precise predictions.

In addition, the adaptability of AI models is a major advantage. Unlike traditional models, which often rely on fixed assumptions, machine learning models can continuously learn from new data and adjust to changing economic conditions.

This makes them particularly useful in volatile markets where financial conditions can shift rapidly. However, the study also highlights several challenges associated with the use of AI and bankruptcy prediction. One of the main issues is the dependence on data quality. Inaccurate, incomplete, or biased data can significantly affect the performance of the models.

Another challenge is the complexity of certain AI techniques, which can make the results difficult to interpret. This lack of transparency, often referred to as the “black box” problem, may limit the practical application of these models, especially in situations where clear explanations are required. Despite these limitations, the overall results suggest that AI offers a valuable enhancement to traditional financial analysis. By combining advanced computational techniques with financial expertise, it is possible to achieve more accurate predictions and gain deeper insights into the factors that contribute to corporate failure.

Discussion and conclusion: "The so what ? "

In conclusion, this study demonstrates that artificial intelligence has the potential to significantly improve the process of bankruptcy prediction. By leveraging financial statement data and applying machine learning techniques, it is possible to identify early warning signs of financial distress with greater accuracy and efficiency than traditional methods. This advancement represents an important step forward in the field of financial risk management, as it enables investors, financial institutions, and other stakeholders make more informed and timely decisions. The integration of AI into financial analysis not only enhances predictive performance but also supports a more proactive approach to risk management.

Instead of reacting to financial problems after they occur, organizations can use these tools to anticipate potential issues and take preventive measures. This contributes to greater financial stability and helps reduce the negative impact of corporate failure on the economy.

However, this study is not without its limitations. One of the primary weaknesses is the reliance on the availability and quality of financial data. If the data used are incomplete outdated or inaccurate, the predictions generated by the models may be unreliable. Additionally, while AI models are powerful, they can sometimes lack transparency, making it difficult for users to fully understand how decisions are made. This can be a significant drawback in areas where accountability and interpretability are essential.

Another limitation is that the study may not fully capture all economic conditions or industry specific factors. Different industries may exhibit unique financial characteristics, and models trained on one data set may not perform equally well in another context. This raises concern about the generalizability of the results.

For future research, it would be beneficial to expand the data set to include a wider range of companies, industries, and economic conditions. Incorporating additional variables such as macroeconomic indicators could also improve the accuracy of predictions.

Furthermore, combining multiple AI techniques, such as hybrid models, may enhance performance and address some of the limitations identified. Despite these weaknesses, the study provides valuable insights into the growing role of artificial intelligence and finance.

It highlights the potential of AI to transform traditional approaches and address real-world financial challenges more effectively.

As technology continues to evolve, its integration into financial risk management will likely become increasingly important, shaping the future of the industry.

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